SpectraViewer

User Manual

Version 1.1.5

English edition, screenshot-free draft

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This manual describes the source-code version of SpectraViewer 1.1.5. A screenshot-enhanced edition can be prepared later from the English user interface.

Contents

1. Introduction

SpectraViewer is a lightweight desktop application for opening, browsing, and visually inspecting spectrum files. It is designed as a fast spectrum browser rather than a full spectrum-processing or spectrum-manipulation environment.

The program can collect supported spectrum files from a folder, from a folder tree, from a manually selected set of files, or from drag-and-drop input, then display them as a navigable list. This makes it useful for quick inspection, checking, and visual comparison of larger spectrum collections.

SpectraViewer does not replace OPUS or other dedicated spectroscopy software. It does not perform spectral processing, manipulation, identification, or database searching. Its purpose is quick viewing and convenient browsing.

1.1 Purpose of this manual

This manual explains the basic operating logic of SpectraViewer 1.1.5, the user interface elements, the supported file-loading workflows, and the practical limitations that users should be aware of. This edition is intentionally text-only; screenshots can be added later after the final English interface screenshots have been prepared.

2. Supported spectrum formats

SpectraViewer currently supports the following file types:

- OPUS-like native spectrum files with numeric extensions, such as .0, .1, .2, and other numeric extensions.
- JCAMP-DX spectrum files: .dx and .jdx.
- DPT data point table files: .dpt.

During file opening, folder scanning, or drag-and-drop operations, the program considers only supported spectrum files. Unsupported files are ignored, and a warning may be displayed when appropriate.

Not all supported file formats contain the same type or amount of metadata. Some files may provide only the spectrum data, while others may also contain sample name, sample form, or other metadata. If metadata fields are missing, the corresponding header lines remain empty.

3. Operating concept

The operating logic of SpectraViewer differs from the traditional “open one file, display one file” approach. The program is primarily designed for browsing through spectrum collections without repeatedly opening individual files.

In normal mode, opening a single supported spectrum file usually creates a browsable list from the surrounding folder context. The opened file becomes the current item in that list, and the user can move to other spectra using the Previous and Next controls or the corresponding keyboard shortcuts.

If Include subfolders is disabled, SpectraViewer builds the list from the supported spectrum files found directly in the folder that contains the opened file. If Include subfolders is enabled, the program scans the folder tree recursively and builds a single navigable list from all supported spectra found under that folder.

This design is especially useful when the goal is not the detailed analysis of a single file, but the fast visual inspection, checking, or comparison of a larger spectrum set. In this sense, SpectraViewer behaves more like a spectrum browser than a conventional spectrum viewer.

3.1 Main browsing modes

- Folder context mode: opening a file or folder builds a list from the selected folder level.
- Recursive folder context mode: opening a file or folder builds a list from the selected folder and its subfolders.
- Custom list mode: selecting or dropping multiple files creates an ad-hoc playlist from the selected supported files.
- Single file mode: only one selected spectrum is loaded, without collecting the surrounding folder context.

4. System requirements and installation

4.1 Source-code version

The GitHub release provides the source-code version of SpectraViewer. It is a Python/Tkinter application and is primarily developed and tested on Windows. The core program should remain portable as long as the required Python packages and Tcl/Tk support are available.

4.2 Main dependencies

- Python 3.
- NumPy.
- Matplotlib.
- Tkinter / Tcl-Tk.
- tkinterdnd2, optional but recommended for drag-and-drop support.
- brukeropus, optional; required for reading OPUS-like native files.

Tkinter is normally included with standard Python installations on Windows. On some Linux distributions it may need to be installed separately through the system package manager.

4.3 Typical installation workflow

From the project folder, a typical Python virtual environment workflow is:

```
python -m venv .venv

.venv\Scripts\activate

pip install -r requirements.txt
```

On Linux or macOS, activate the virtual environment with the appropriate shell command instead.

4.4 Running the program

From the project folder, run:

```
python SpectraViewer.py
```

5. User interface overview

The SpectraViewer interface is intentionally simple. The upper part of the window contains the main file and folder controls, the central area displays the current spectrum, and the lower part contains the Matplotlib navigation toolbar and the status bar.

5.1 Main controls

Control	Function
Path field	Shows the current path or folder context. It can also be used to type or paste a file or folder path and open it with Enter.
Folder…	Opens a folder and builds a navigable list from supported spectrum files found there. If Include subfolders is enabled, subfolders are scanned as well.
File…	Opens one or more supported spectrum files. In normal mode, opening one file builds a folder-context list; selecting multiple files creates a custom list.
Include subfolders	Controls whether folder scanning is limited to the current folder or includes the full folder tree below it.
Single file	Loads only one selected spectrum without collecting surrounding files. While enabled, Include subfolders is disabled temporarily.
About	Displays version information, supported formats, and keyboard shortcuts.
Previous / Next	Moves through the active file list. These buttons are disabled automatically when the active list contains fewer than two spectra.
Save image	Exports the currently displayed plot as an image file.
Color	Changes the display color of the spectrum curve.
Invert X	Controls the display direction of the x-axis. For FTIR-style wavenumber display, the axis is usually shown from high to low wavenumber.
Crosshair	Displays coordinate-reading helper lines in the plot area. The crosshair can also be toggled with the right mouse button inside the plot area.

5.2 Spectrum display area

The central plot area shows the currently selected spectrum. The header above the plot displays the file name and, when available, the most relevant metadata fields such as sample name and sample form. If a file does not contain these metadata fields, the corresponding lines remain empty.

When the crosshair is enabled, the cursor position is followed by horizontal and vertical helper lines. The current coordinates are displayed in the plot area and can be useful for approximate peak position reading or quick visual checking.

5.3 Matplotlib toolbar and status bar

The Matplotlib toolbar below the plot provides standard viewing tools such as home view, pan, zoom, and subplot adjustment. SpectraViewer removes the default Matplotlib save button from the toolbar and provides its own Save image function instead.

The status bar shows short feedback messages, such as the active file path, save status, warnings, or error information.

6. Loading files and folders

6.1 Opening a folder

Use Folder… to select a directory. SpectraViewer scans the selected directory for supported spectrum files and builds a navigable list. If Include subfolders is active, the scan is recursive.

6.2 Opening one file

In normal mode, opening a single supported file does not necessarily create a one-file view. Instead, SpectraViewer uses the file as an entry point into its folder context and builds a list from the surrounding supported files. The selected file becomes the current position in that list.

6.3 Opening multiple files

When multiple files are selected in the file dialog, SpectraViewer creates a custom list from the selected supported files. Unsupported items are skipped, and a warning may be displayed if unsupported files were selected.

6.4 Typing or pasting a path

The Path field can be used to type or paste a file or folder path. Press Enter to open it. If the path is a folder, the folder is scanned. If the path is a supported spectrum file, SpectraViewer applies the current loading logic: folder context in normal mode, or one-file loading in Single file mode.

7. Single file mode

Single file mode was introduced in version 1.1.5 to provide a clear, intentional alternative to the normal folder-context browsing logic. It is useful when the user wants to inspect exactly one spectrum without collecting neighboring files.

When Single file mode is enabled, the Include subfolders option is temporarily disabled. SpectraViewer preserves the previous Include subfolders state internally. If a spectrum is already loaded, the active list is reduced to the current file and the counter changes to 1/1.

While Single file mode is active:

- Opening one supported file loads only that file.
- Dropping one supported file loads only that file.
- Dropping or selecting multiple files is rejected with a warning.
- Opening a folder is not allowed, because folder loading contradicts the single-file logic.
- Previous and Next navigation controls are disabled because there is no multi-file list to browse.

When Single file mode is turned off, SpectraViewer restores the previous Include subfolders setting and attempts to rebuild the folder context from the current spectrum. If the restored list contains at least two spectra, navigation becomes available again.

8. Navigation and plot controls

8.1 Spectrum navigation

The Previous and Next buttons move through the active list. Navigation wraps around at the beginning and end of the list. The Home and End keys jump to the first and last item, while PageUp and PageDown jump by larger steps.

Version 1.1.5 disables Previous and Next automatically when the active list contains zero or one spectrum. This avoids misleading navigation controls when there is nothing to browse.

8.2 X-axis direction

The Invert X checkbox controls whether the x-axis is displayed in reversed order. This is useful for FTIR-style wavenumber display, where spectra are commonly shown from higher to lower wavenumber.

8.3 Crosshair

The Crosshair option displays horizontal and vertical helper lines that follow the cursor in the plot area. It can be toggled by the checkbox or by right-clicking inside the spectrum display area. If the crosshair is visible when saving the plot, it will be included in the exported image.

8.4 Spectrum color

The Color button opens a color chooser and changes the line color of the displayed spectrum. This can be useful for preparing quick illustrations or for adapting exported images to a specific document style.

9. Drag and drop

If tkinterdnd2 is available, SpectraViewer accepts drag-and-drop input from the file manager. Users can drop one or more files, folders, or mixed selections onto the application window.

A single dropped file follows the same logic as a typed path or file-dialog selection. In normal mode, it loads the folder context. In Single file mode, it loads only the dropped file.

Multiple dropped items are treated as an ad-hoc playlist in normal mode. Supported files are collected, folders are expanded according to the current Include subfolders setting, duplicates are removed, and unsupported files are skipped. In Single file mode, dropping multiple items is rejected with a warning.

This behavior is useful when files are selected manually from different folders or from a Windows Explorer search result and then dropped into SpectraViewer as a temporary browsing set.

10. Exporting plots

Use Save image to export the currently displayed spectrum plot. Supported output formats include PNG, JPEG, PDF, and SVG, depending on the selected file extension and Matplotlib support.

The saved image reflects the current visible plot state. If the crosshair is enabled and visible, it is included in the exported image. If the crosshair is disabled, it is not included.

11. Notes, limitations, and file-count safeguards

11.1 Not a processing tool

SpectraViewer is intended for viewing and browsing. It does not apply baseline correction, normalization, smoothing, spectral identification, database searching, or other analytical processing steps.

SpectraViewer displays a single spectrum at a time. It does not provide stacked or overlaid multi-spectrum views, as commonly used in dedicated spectral evaluation or processing software.

11.2 OPUS file complexity

Some OPUS files can contain multiple data blocks, such as single-channel data, reference data, mapping points, images, or other auxiliary information. In such cases, SpectraViewer may not always display the spectrum that the user expects as the primary spectrum. If a complex OPUS file does not display correctly, extracting the desired spectrum within OPUS and opening the simpler extracted file may help.

11.3 Metadata differences

Different supported formats store metadata differently. Some files contain sample names, sample forms, or labels; others contain only x/y data. SpectraViewer displays available metadata when it can read it, but the absence of metadata is not treated as an error.

11.4 File-count safeguards

To avoid excessive memory use and very slow operation, SpectraViewer checks the number of files before replacing the active list. Large lists may trigger a warning. If a scan exceeds the configured maximum, the operation is cancelled and the previous valid list is preserved.

Version 1.1.4 improved this behavior so that rejected scans no longer overwrite the previously loaded safe state. Version 1.1.5 keeps this behavior and extends the navigation state logic to zero- and one-file lists.

12. Citation, DOI, and license

SpectraViewer 1.1.5 is archived on Zenodo and can be cited as research software.

Version DOI: 10.5281/zenodo.20715099

The GitHub repository also contains a CITATION.cff file so that GitHub can display citation information for the project.

SpectraViewer is released under the MIT License. The license text is included in the repository as LICENSE.

SpectraViewer was developed by Zsolt Hidasi with assistance from OpenAI ChatGPT during design, refactoring, documentation, and release preparation.

Appendix A. Keyboard shortcuts

Shortcut	Action
Ctrl+O	Open file
Ctrl+Shift+O	Open folder
Left / Right	Previous / Next spectrum
Home / End	First / Last spectrum

PageUp / PageDown	Jump backward / forward through the active list
Ctrl+S	Save current plot as image
Ctrl+Shift+C	Change spectrum color
F1	Open the About window

Appendix B. Version 1.1.5 change summary

- Added Single file mode for opening and viewing only one selected spectrum without loading the surrounding folder context.
- Single file mode is applied consistently to file dialog, typed path, and drag-and-drop workflows.
- When Single file mode is enabled, subfolder traversal is temporarily disabled while preserving its previous state.
- When Single file mode is disabled, the previous folder context is rebuilt from the current spectrum using the restored subfolder traversal setting.
- Previous/Next navigation controls are disabled automatically when there are fewer than two spectra in the active list.
- User interface text has been moved to a centralized UI_TEXT dictionary to make localization cleaner and easier to maintain.
- Version 1.1.5 is preserved as a functional milestone with matching English and Hungarian UI variants.